Deploy the kube-prometheus-stack Using Helm

Prerequisites

```
nelesawy@localhost:~$ kubectl config current-context
docker-desktop
nelesawy@localhost:~$ kubectl get nodes
NAME                    STATUS   ROLES           AGE     VERSION
desktop-control-plane   Ready    control-plane   3h49m   v1.31.1
desktop-worker          Ready    <none>          3h49m   v1.31.1
nelesawy@localhost:~$ helm version
version.BuildInfo{Version:"v3.21.2", GitCommit:"125963406833fe0525be91f46c8b5b0f22fb9e32", GitTreeState:"clean", GoVersion:"go1.26.4"}
nelesawy@localhost:~$
```

Task 1 – Configure Helm

```
nelesawy@localhost:~$ helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
"prometheus-community" has been added to your repositories
nelesawy@localhost:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "prometheus-community" chart repository
Update Complete. *Happy Helming!*
nelesawy@localhost:~$ helm repo list
NAME                    URL
prometheus-community    https://prometheus-community.github.io/helm-charts
nelesawy@localhost:~$ helm search repo prometheus-community/kube-prometheus-stack
NAME                                            CHART VERSION   APP VERSION     DESCRIPTION
prometheus-community/kube-prometheus-stack      87.2.1          v0.92.0         kube-prometheus-stack collects Kubernetes manif...
```

Task 2 – Prepare the Namespace

```
nelesawy@localhost:~$ kubectl create namespace monitoring
namespace/monitoring created
nelesawy@localhost:~$ kubectl get namespaces
NAME                 STATUS   AGE
default              Active   96m
kube-node-lease      Active   96m
kube-public          Active   96m
kube-system          Active   96m
local-path-storage   Active   96m
monitoring           Active   12s
```

Task 3 – Deploy the Stack

```
nelesawy@localhost:~$ helm install monitoring-stack  prometheus-community/kube-prometheus-stack  --namespace monitoring   --version 87.2.1   --wait   --timeou
t 10m
NAME: monitoring-stack
LAST DEPLOYED: Sat Jun 27 15:10:56 2026
NAMESPACE: monitoring
STATUS: deployed
REVISION: 1
TEST SUITE: None
```

```
nelesawy@localhost:~$ helm list -n monitoring
NAME                    NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                           APP VE
RSION
monitoring-stack        monitoring      1               2026-06-27 15:10:56.064260258 +0300 EEST        deployed        kube-prometheus-stack-87.2.1    v0.92.
0
```

Task 4 – Verify the Installation

```
nelesawy@localhost:~$ kubectl get pods,deployments,statefulsets,daemonsets,services,configmaps,secrets,serviceaccounts -n monitoring
NAME                                                         READY   STATUS    RESTARTS   AGE
pod/alertmanager-monitoring-stack-kube-prom-alertmanager-0   2/2     Running   0          80m
pod/monitoring-stack-grafana-5649fff595-5njtc                3/3     Running   0          83m
pod/monitoring-stack-kube-prom-operator-5d4874d5db-pz98h     1/1     Running   0          83m
pod/monitoring-stack-kube-state-metrics-76d8c6f694-s8xfz     1/1     Running   0          83m
pod/monitoring-stack-prometheus-node-exporter-6qkjr          1/1     Running   0          83m
pod/monitoring-stack-prometheus-node-exporter-g7rxj          1/1     Running   0          83m
pod/prometheus-monitoring-stack-kube-prom-prometheus-0       2/2     Running   0          80m

NAME                                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/monitoring-stack-grafana              1/1     1            1           83m
deployment.apps/monitoring-stack-kube-prom-operator   1/1     1            1           83m
deployment.apps/monitoring-stack-kube-state-metrics   1/1     1            1           83m

NAME                                                                    READY   AGE
statefulset.apps/alertmanager-monitoring-stack-kube-prom-alertmanager   1/1     80m
statefulset.apps/prometheus-monitoring-stack-kube-prom-prometheus       1/1     80m

NAME                                                       DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR            AGE
daemonset.apps/monitoring-stack-prometheus-node-exporter   2         2         2       2            2           kubernetes.io/os=linux   83m
```

```
NAME                                                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)                      AGE
service/alertmanager-operated                        ClusterIP   None            <none>        9093/TCP,9094/TCP,9094/UDP   80m
service/monitoring-stack-grafana                     ClusterIP   10.96.151.87    <none>        80/TCP                       83m
service/monitoring-stack-kube-prom-alertmanager      ClusterIP   10.96.51.188    <none>        9093/TCP,8080/TCP            83m
service/monitoring-stack-kube-prom-operator          ClusterIP   10.96.218.152   <none>        443/TCP                      83m
service/monitoring-stack-kube-prom-prometheus        ClusterIP   10.96.71.60     <none>        9090/TCP,8080/TCP            83m
service/monitoring-stack-kube-state-metrics          ClusterIP   10.96.184.16    <none>        8080/TCP                     83m
service/monitoring-stack-prometheus-node-exporter    ClusterIP   10.96.49.2      <none>        9100/TCP                     83m
service/prometheus-operated                          ClusterIP   None            <none>        9090/TCP                     80m
```

```
NAME                                                                               TYPE                 DATA   AGE
secret/alertmanager-monitoring-stack-kube-prom-alertmanager                        Opaque               1      83m
secret/alertmanager-monitoring-stack-kube-prom-alertmanager-cluster-tls-config     Opaque               1      80m
secret/alertmanager-monitoring-stack-kube-prom-alertmanager-generated              Opaque               1      80m
secret/alertmanager-monitoring-stack-kube-prom-alertmanager-tls-assets-0           Opaque               0      80m
secret/alertmanager-monitoring-stack-kube-prom-alertmanager-web-config             Opaque               1      80m
secret/monitoring-stack-grafana                                                    Opaque               3      83m
secret/monitoring-stack-kube-prom-admission                                        Opaque               3      83m
secret/prometheus-monitoring-stack-kube-prom-prometheus                            Opaque               1      80m
secret/prometheus-monitoring-stack-kube-prom-prometheus-thanos-prometheus-http-client-file   Opaque     1      80m
secret/prometheus-monitoring-stack-kube-prom-prometheus-tls-assets-0               Opaque               1      80m
secret/prometheus-monitoring-stack-kube-prom-prometheus-web-config                 Opaque               1      80m
secret/sh.helm.release.v1.monitoring-stack.v1                                      helm.sh/release.v1   1      84m
```

```
NAME                                                                   DATA   AGE
configmap/kube-root-ca.crt                                             1      141m
configmap/monitoring-stack-grafana                                     1      83m
configmap/monitoring-stack-grafana-config-dashboards                   1      83m
configmap/monitoring-stack-kube-prom-alertmanager-overview             1      83m
configmap/monitoring-stack-kube-prom-apiserver                         1      83m
configmap/monitoring-stack-kube-prom-cluster-total                     1      83m
configmap/monitoring-stack-kube-prom-controller-manager                1      83m
configmap/monitoring-stack-kube-prom-etcd                              1      83m
configmap/monitoring-stack-kube-prom-grafana-datasource                1      83m
configmap/monitoring-stack-kube-prom-grafana-overview                  1      83m
configmap/monitoring-stack-kube-prom-k8s-coredns                       1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-cluster             1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-multicluster        1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-namespace           1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-node                1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-pod                 1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-workload            1      83m
configmap/monitoring-stack-kube-prom-k8s-resources-workloads-namespace 1      83m
configmap/monitoring-stack-kube-prom-kubelet                           1      83m
configmap/monitoring-stack-kube-prom-namespace-by-pod                  1      83m
configmap/monitoring-stack-kube-prom-namespace-by-workload             1      83m
configmap/monitoring-stack-kube-prom-node-cluster-rsrc-use             1      83m
configmap/monitoring-stack-kube-prom-node-rsrc-use                     1      83m
configmap/monitoring-stack-kube-prom-nodes                             1      83m
configmap/monitoring-stack-kube-prom-nodes-aix                         1      83m
configmap/monitoring-stack-kube-prom-nodes-darwin                      1      83m
configmap/monitoring-stack-kube-prom-persistentvolumesusage            1      83m
configmap/monitoring-stack-kube-prom-pod-total                         1      83m
configmap/monitoring-stack-kube-prom-prometheus                        1      83m
configmap/monitoring-stack-kube-prom-proxy                             1      83m
configmap/monitoring-stack-kube-prom-scheduler                         1      83m
configmap/monitoring-stack-kube-prom-workload-total                    1      83m
configmap/prometheus-monitoring-stack-kube-prom-prometheus-rulefiles-0 35     80m
```

Task 5 – Identify the Stack Components

```
NAME                                                   SECRETS   AGE
serviceaccount/default                                 0         141m
serviceaccount/monitoring-stack-grafana                0         83m
serviceaccount/monitoring-stack-kube-prom-alertmanager 0         83m
serviceaccount/monitoring-stack-kube-prom-operator     0         83m
serviceaccount/monitoring-stack-kube-prom-prometheus   0         83m
serviceaccount/monitoring-stack-kube-state-metrics     0         83m
serviceaccount/monitoring-stack-prometheus-node-exporter 0       83m
nelesawy@localhost:~$
```

```
nelesawy@localhost:~$ kubectl get pods -n monitoring
NAME                                                   READY   STATUS    RES
alertmanager-monitoring-stack-kube-prom-alertmanager-0 2/2     Running   0
monitoring-stack-grafana-5649fff595-5njtc              3/3     Running   0
monitoring-stack-kube-prom-operator-5d4874d5db-pz98h   1/1     Running   0
monitoring-stack-kube-state-metrics-76d8c6f694-s8xfz   1/1     Running   0
monitoring-stack-prometheus-node-exporter-6qkjr        1/1     Running   0
monitoring-stack-prometheus-node-exporter-g7rxj        1/1     Running   0
prometheus-monitoring-stack-kube-prom-prometheus-0     2/2     Running   0
```

1-

```
prometheus-monitoring-stack-kube-prom-prometheus-0      2/2     Running   0
```

Prometheus Server: (watch cluster)
It collects, stores, and queries metrics from Kubernetes targets using PromQL.

2-

```
monitoring-stack-grafana-5649fff595-5njtc               3/3     Running   0
```

Grafana: (display graphs of matrics)
Grafana provides dashboards and visualizations for metrics collected by Prometheus.

3-

```
alertmanager-monitoring-stack-kube-prom-alertmanager-0   2/2     Running   0
```

Alertmanager:(manage alert , send emails)
It receives alerts from Prometheus and manages alert routing, grouping, and notifications to user.

4-

```
monitoring-stack-kube-state-metrics-76d8c6f694-s8xfz     1/1     Running   0
```

kube-state-metrics:(watch k8s obj)
It exposes metrics about Kubernetes object states such as pods, deployments, replicas, nodes, and workloads.

5-

```
monitoring-stack-kube-prom-operator-5d4874d5db-pz98h     1/1     Running   0
```

Prometheus Operator:(manage prom stack inside cluster)
It manages Prometheus, Alertmanager, ServiceMonitor, PodMonitor, and PrometheusRule resources in a Kubernetes-native way.

6-

```
monitoring-stack-prometheus-node-exporter-6qkjr     1/1     Running   0     6h3m   172.18.0.2   desktop-control-plane
monitoring-stack-prometheus-node-exporter-g7rxj     1/1     Running   0     6h3m   172.18.0.3   desktop-worker
```

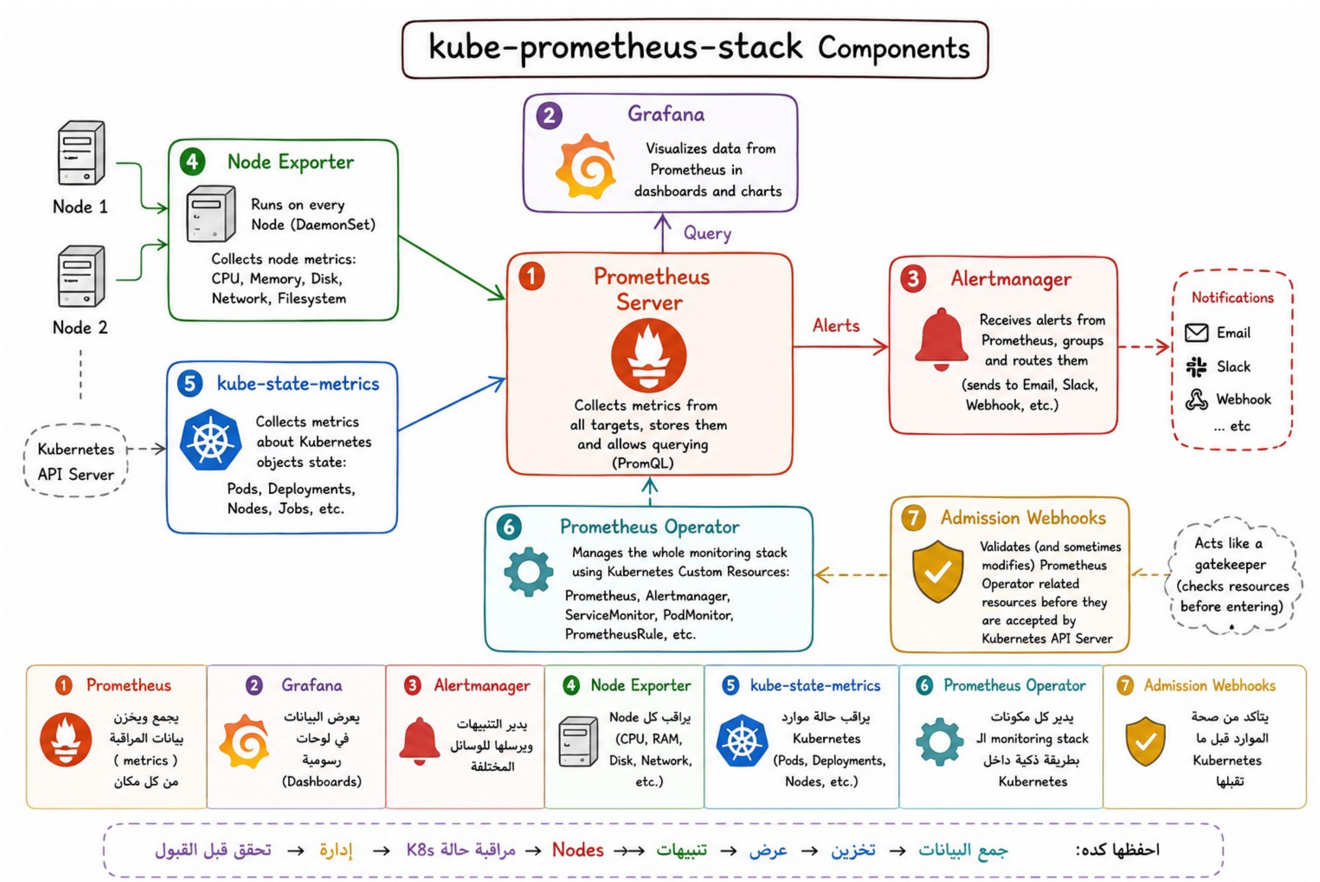
Node Exporter:(watch server)
Node Exporter is deployed as a DaemonSet.

It runs one pod per node and collects node-level metrics such as CPU, memory, disk, filesystem, and network usage.

7-

```
nelesawy@localhost:~$ kubectl get validatingwebhookconfigurations | grep monitoring
monitoring-stack-kube-prom-admission   2          5h55m
```

Admission Webhooks:(check resources before join cluster)
Admission webhooks are installed by the Prometheus Operator chart to validate or manage Prometheus-related Kubernetes resources before they are accepted by the Kubernetes API server.

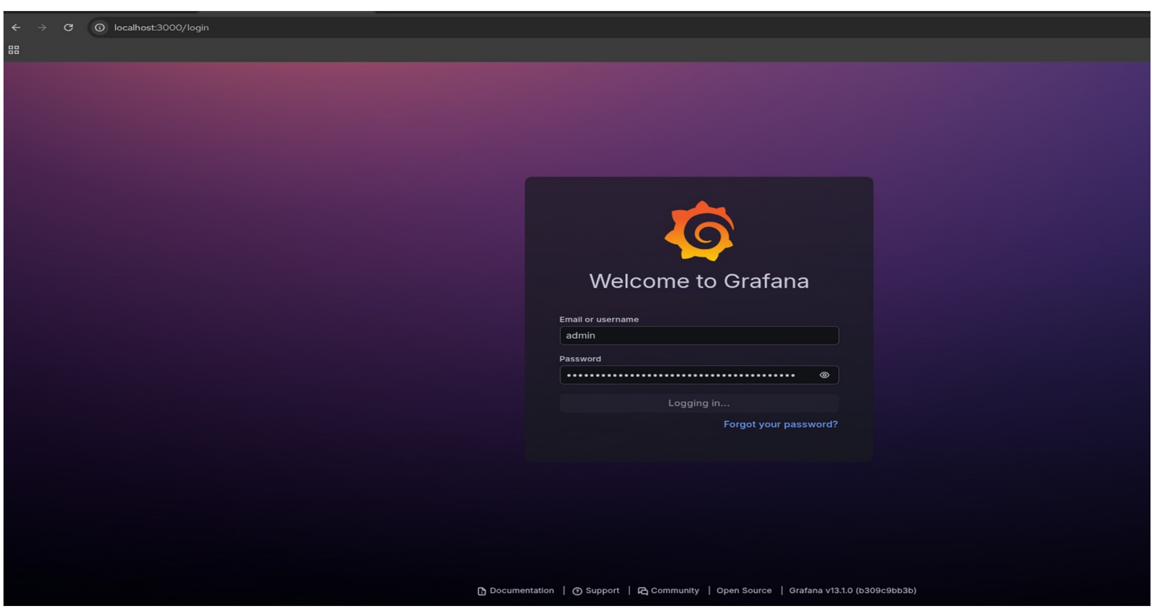


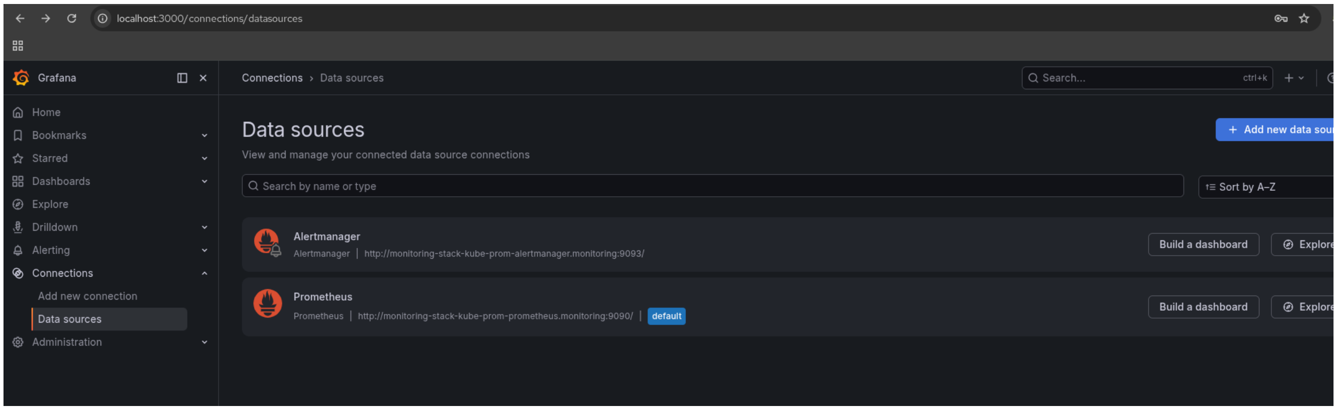
Task 6 – Explore the Helm Release

```
nelesawy@localhost:~$ helm list -n monitoring -o yaml
- app_version: v0.92.0
  chart: kube-prometheus-stack-87.2.1
  name: monitoring-stack
  namespace: monitoring
  revision: "1"
  status: deployed
  updated: 2026-06-27 15:10:56.064260258 +0300 EEST
nelesawy@localhost:~$
```

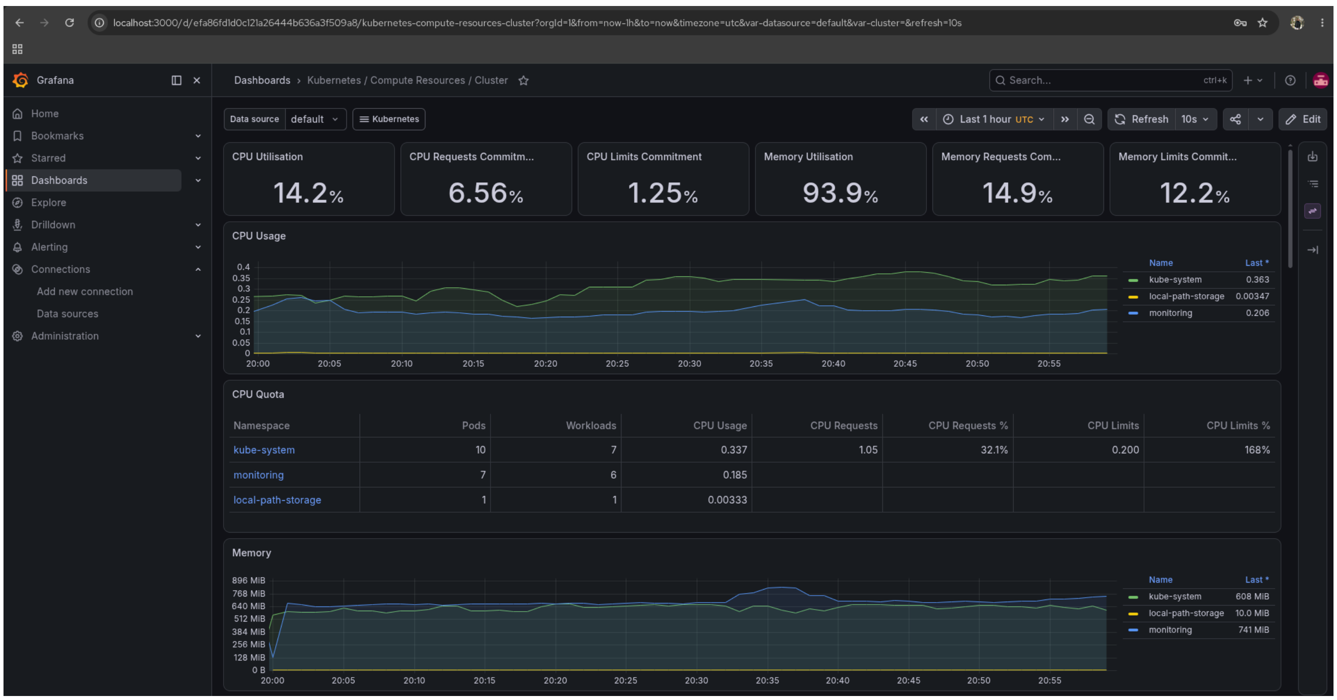
Task 7 – Access Grafana

```
nelesawy@localhost:~$ kubectl --namespace monitoring get secrets monitoring-stack-grafana \
  -o jsonpath="{.data.admin-password}" | base64 -d ; echo
BDY0tVmKWGfK8kQDZP2PZ8SiSptARtquvWVddGGZ
nelesawy@localhost:~$ kubectl -n monitoring port-forward svc/monitoring-stack-grafana 3000:80
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
Handling connection for 3000
```





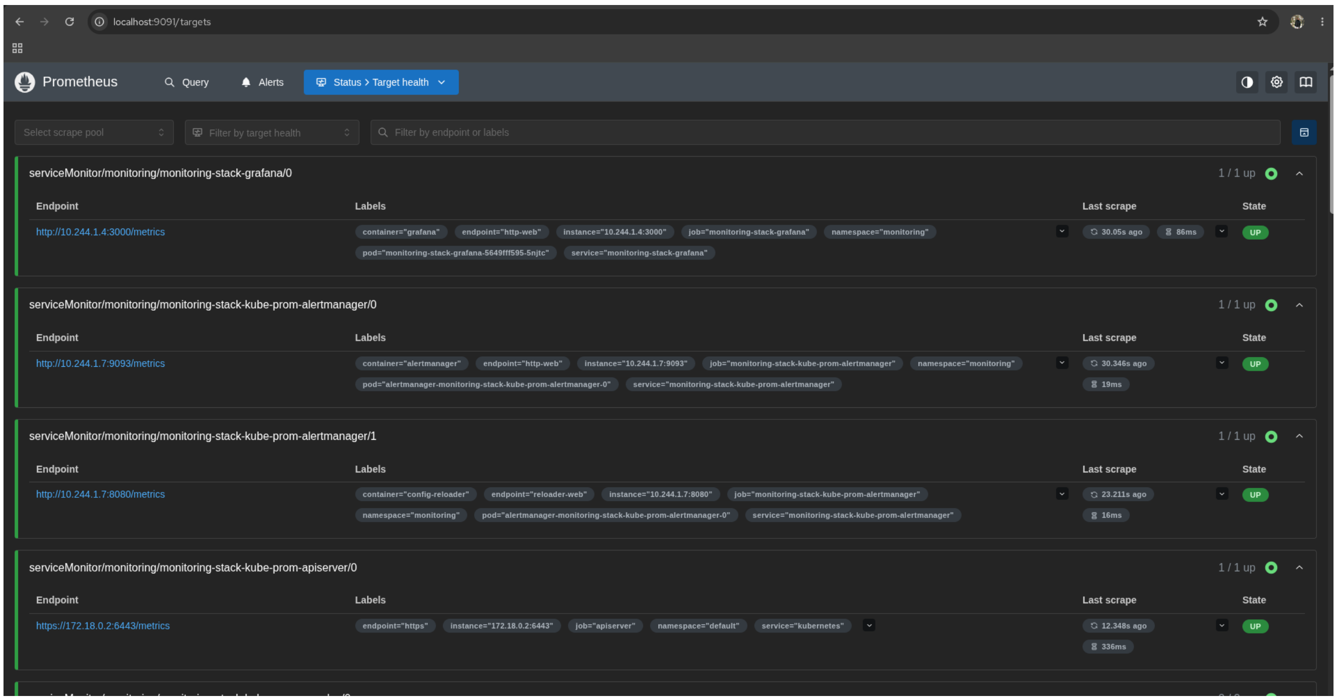
localhost:3000/connections/datasources
Grafana
Connections › Data sources
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Data sources
View and manage your connected data source connections
Search by name or type
Sort by A–Z
Alertmanager
Alertmanager | http://monitoring-stack-kube-prom-alertmanager.monitoring:9093/
Build a dashboard
Prometheus
Prometheus | http://monitoring-stack-kube-prom-prometheus.monitoring:9090/ | default
Build a dashboard



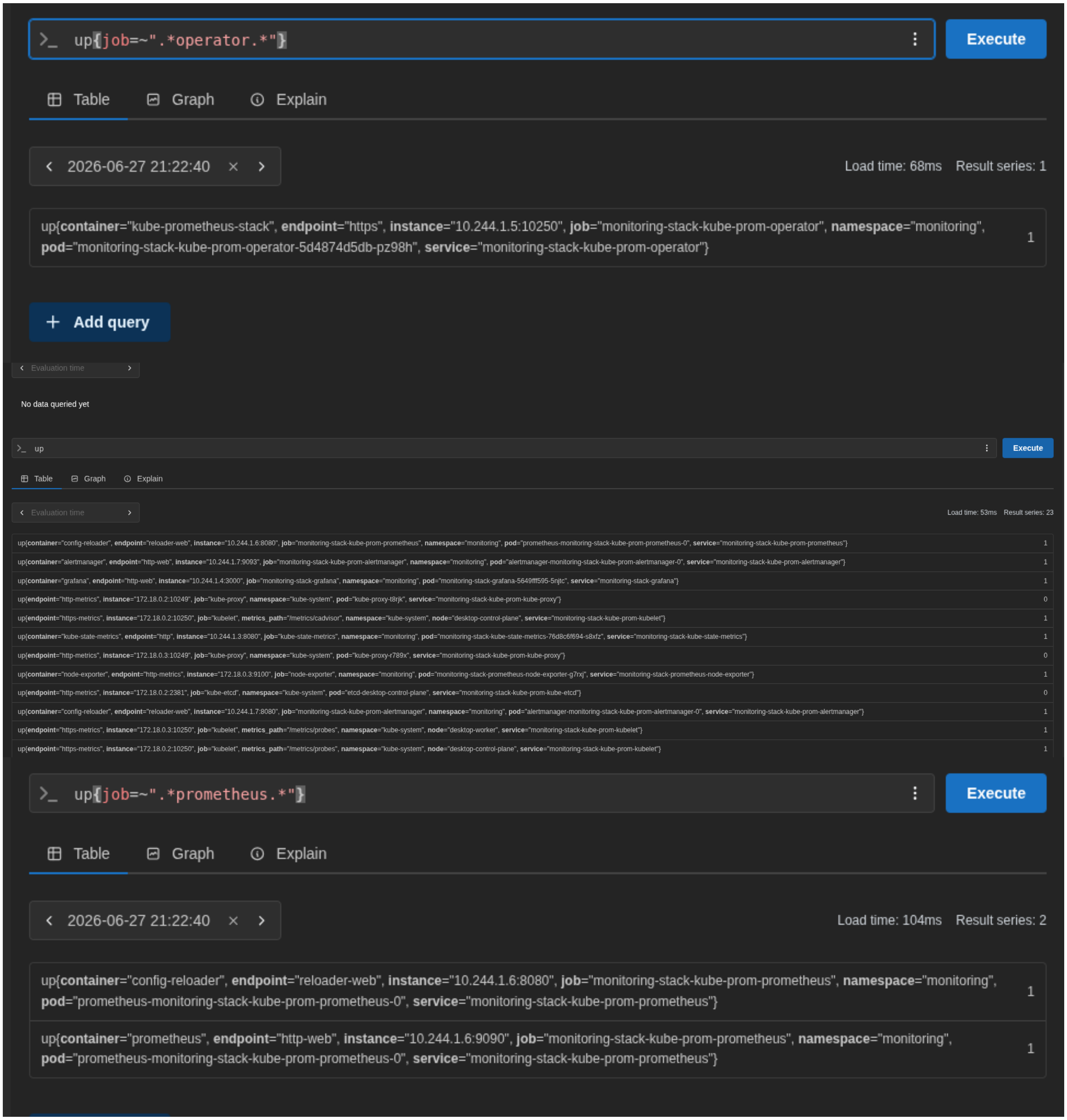
localhost:3000/d/efa86fd1d0c121a26444b636a3f509a8/kubernetes-compute-resources-cluster?orgId=1&from=now-1h&to=now&timezone=utc&var-datasource=default&var-cluster=&refresh=10s
Grafana
Dashboards › Kubernetes / Compute Resources / Cluster
Data source default
Kubernetes
Last 1 hour UTC
Refresh 10s
Edit
CPU Utilisation 14.2%
CPU Requests Commitm... 6.56%
CPU Limits Commitment 1.25%
Memory Utilisation 93.9%
Memory Requests Com... 14.9%
Memory Limits Commit... 12.2%
CPU Usage
kube-system 0.363
local-path-storage 0.00347
monitoring 0.206
CPU Quota
Namespace | Pods | Workloads | CPU Usage | CPU Requests | CPU Requests % | CPU Limits | CPU Limits %
kube-system | 10 | 7 | 0.337 | 1.05 | 32.1% | 0.200 | 168%
monitoring | 7 | 6 | 0.185
local-path-storage | 1 | 1 | 0.00333
Memory
kube-system 608 MiB
local-path-storage 10.0 MiB
monitoring 741 MiB

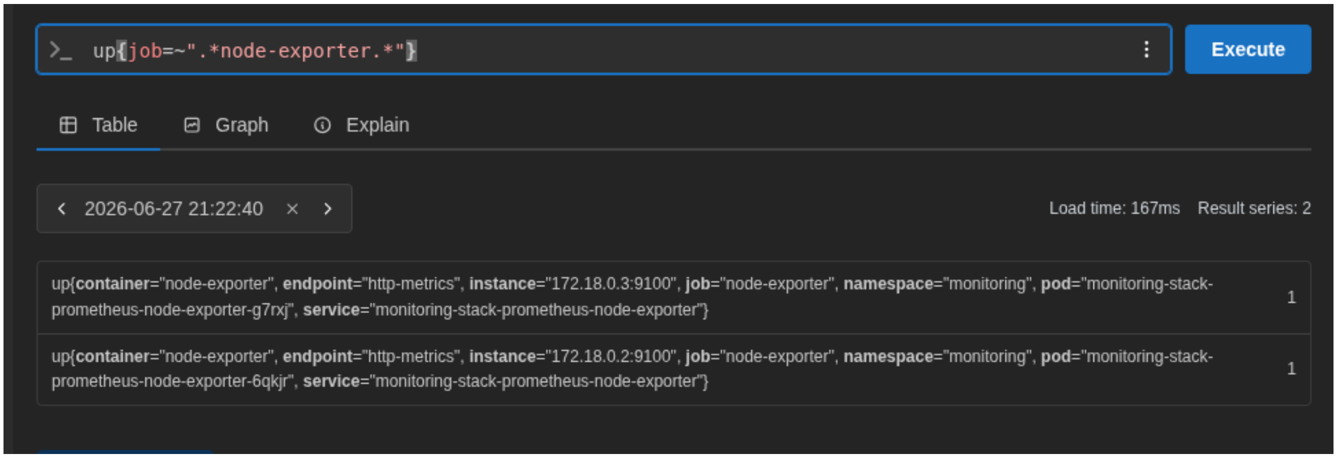
Task 8 – Access Prometheus

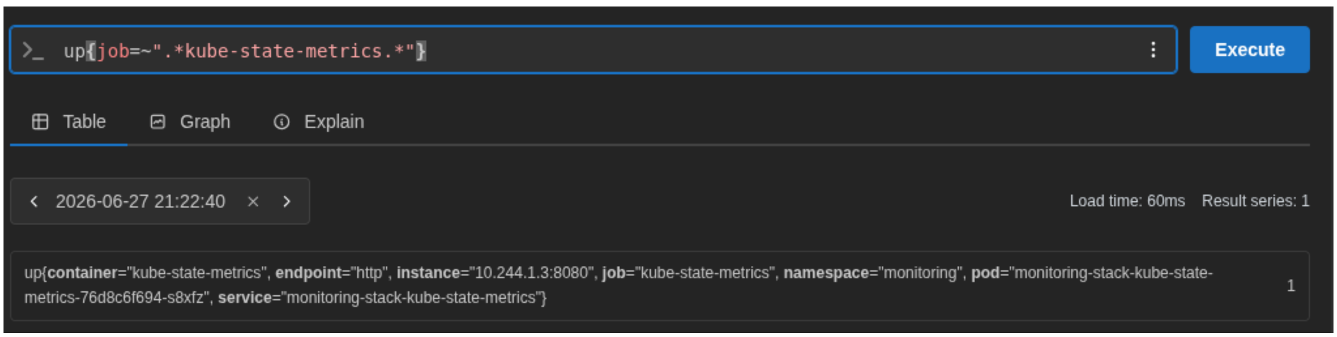
```
nelesawy@localhost:~$ kubectl -n monitoring port-forward svc/monitoring-stack-kube-prom-prometheus 9091:9090
Forwarding from 127.0.0.1:9091 -> 9090
Forwarding from [::1]:9091 -> 9090
Handling connection for 9091
Handling connection for 9091
Handling connection for 9091
Handling connection for 9091
Handling connection for 9091
Handling connection for 9091
```



Task 9 – Verify Monitoring Targets







Kubernetes API Server: UP
kubelet: UP
CoreDNS: UP
kube-state-metrics: UP
Node Exporter: UP
Prometheus Operator: UP
Prometheus Server: UP

Task 10 – Explore Custom Resources (CRD)

```
nelesawy@localhost:~$ kubectl get crds | grep monitoring.coreos.com
alertmanagerconfigs.monitoring.coreos.com   2026-06-27T12:10:53Z
alertmanagers.monitoring.coreos.com         2026-06-27T12:10:53Z
podmonitors.monitoring.coreos.com           2026-06-27T12:10:54Z
probes.monitoring.coreos.com                2026-06-27T12:10:54Z
prometheusagents.monitoring.coreos.com      2026-06-27T12:10:54Z
prometheuses.monitoring.coreos.com          2026-06-27T12:10:54Z
prometheusrules.monitoring.coreos.com       2026-06-27T12:10:54Z
scrapeconfigs.monitoring.coreos.com         2026-06-27T12:10:54Z
servicemonitors.monitoring.coreos.com       2026-06-27T12:10:55Z
thanosrulers.monitoring.coreos.com          2026-06-27T12:10:55Z
nelesawy@localhost:~$
```

```
nelesawy@localhost:~$ kubectl get prometheus,alertmanager,servicemonitor,podmonitor,prometheusrule,probe,thanosruler -n monitoring
NAME                                                              VERSION              DESIRED   READY   RECONCILED   AVAILABLE   AGE
prometheus.monitoring.coreos.com/monitoring-stack-kube-prom-prometheus   v3.12.0-distroless   1         1       True         True        9h

NAME                                                                  VERSION   REPLICAS   READY   RECONCILED   AVAILABLE   AGE
alertmanager.monitoring.coreos.com/monitoring-stack-kube-prom-alertmanager   v0.33.0   1          1       True         True        9h

NAME                                                                              AGE
servicemonitor.monitoring.coreos.com/monitoring-stack-grafana                     9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-alertmanager      9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-apiserver         9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-coredns           9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-kube-controller-manager   9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-kube-etcd         9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-kube-proxy        9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-kube-scheduler    9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-kubelet           9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-operator          9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-prom-prometheus        9h
servicemonitor.monitoring.coreos.com/monitoring-stack-kube-state-metrics          9h
servicemonitor.monitoring.coreos.com/monitoring-stack-prometheus-node-exporter    9h
```

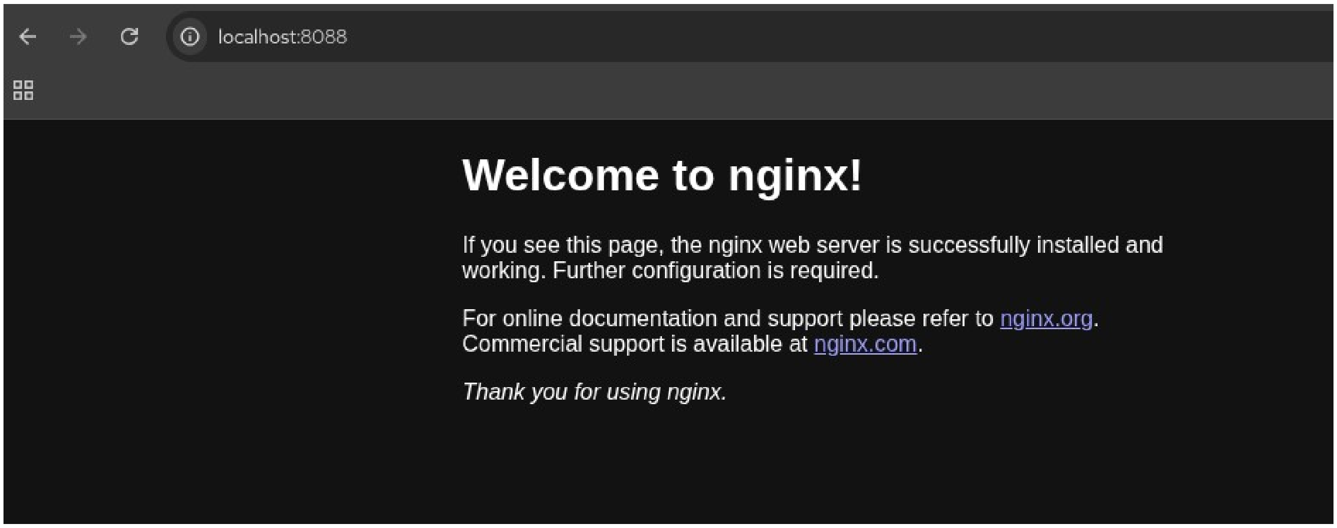
```
NAME                                                                                                  AGE
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-alertmanager.rules                   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-config-reloaders                     9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-etcd                                 9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-general.rules                        9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-cpu-usage-second   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-memory-cache     9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-memory-rss       9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-memory-swap      9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-memory-working-s   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.container-resource         9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-k8s.rules.pod-owner                  9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-apiserver-availability.rules    9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-apiserver-burnrate.rules        9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-apiserver-histogram.rules       9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-apiserver-slos                  9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-prometheus-general.rules        9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-prometheus-node-recording.rules   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-scheduler.rules                 9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kube-state-metrics                   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubelet.rules                        9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-apps                      9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-resources                 9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-storage                   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system                    9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system-apiserver          9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system-controller-manager   9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system-kube-proxy         9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system-kubelet            9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-kubernetes-system-scheduler          9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-node-exporter                        9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-node-exporter.rules                  9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-node-network                         9h
prometheusrule.monitoring.coreos.com/monitoring-stack-kube-prom-node.rules                           9h
```

Task 11 – Deploy an Application

```
nelesawy@localhost:~$ kubectl create namespace sample-app
namespace/sample-app created
nelesawy@localhost:~$ kubectl get ns sample-app
NAME         STATUS   AGE
sample-app   Active   10s
nelesawy@localhost:~$ kubectl create deployment nginx-demo \
  --image=nginx:latest \
  --port=80 \
  -n sample-app
deployment.apps/nginx-demo created
```

```
nelesawy@localhost:~$ kubectl get pods -n sample-app
NAME                          READY   STATUS    RESTARTS   AGE
nginx-demo-76f48575bc-jnv5x   1/1     Running   0          4m20s
nelesawy@localhost:~$
```

```
nelesawy@localhost:~$ kubectl expose deployment nginx-demo \
  --type=ClusterIP \
  --port=80 \
  --target-port=80 \
  -n sample-app
service/nginx-demo exposed
nelesawy@localhost:~$ kubectl get svc -n sample-app
NAME         TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
nginx-demo   ClusterIP   10.96.76.92   <none>        80/TCP    11s
nelesawy@localhost:~$ kubectl -n sample-app port-forward svc/nginx-demo 8088:80
Forwarding from 127.0.0.1:8088 -> 80
Forwarding from [::1]:8088 -> 80
Handling connection for 8088
Handling connection for 8088
```


localhost:8088
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.